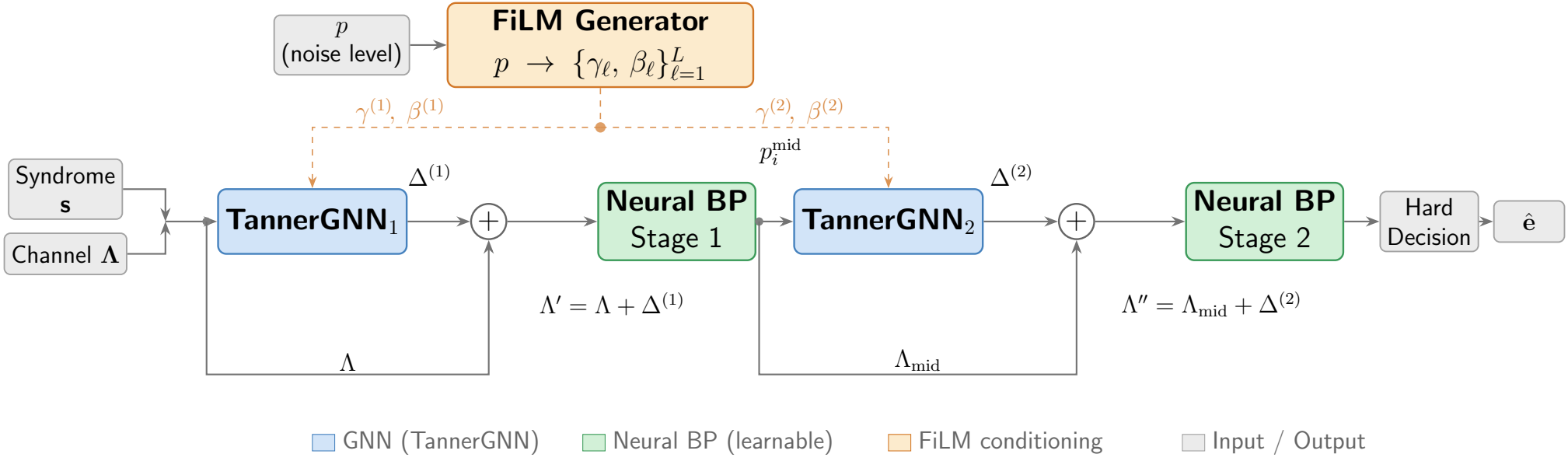

p
(noise level)
FiLM Generator
$p \rightarrow \{\gamma_\ell, \beta_\ell\}_{\ell=1}^{L}$
$\gamma^{(1)}, \beta^{(1)}$
$\gamma^{(2)}, \beta^{(2)}$
p_i^{mid}
Syndrome
s
Channel Λ
TannerGNN$_1$
$\Delta^{(1)}$
Neural BP
Stage 1
$\Lambda' = \Lambda + \Delta^{(1)}$
Λ
TannerGNN$_2$
$\Delta^{(2)}$
Neural BP
Stage 2
$\Lambda'' = \Lambda_{\text{mid}} + \Delta^{(2)}$
Λ_{mid}
Hard
Decision
$\hat{\mathbf{e}}$
GNN (TannerGNN)
Neural BP (learnable)
FiLM conditioning
Input / Output